

Digital Watermarking using Unsigned Integer for Image Authentication

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Abstract— Active tampering detection systems based on watermarking are developing rapidly. This is in line with the increasing number of image manipulation processes that even ordinary people can easily use through various applications that can be used free of charge. The singular value-based watermarking method is very popular because of its good authentication capabilities but requires large computational costs. Several alternative solutions based on signed integer matrices such as Hadamard and Walsh are able to speed up the computational process and produce a good level of imperceptibility. The two matrices have the same characteristics in the first row and column, while the other ones have opposite integer signs. This study proposes the formation of a new matrix based on the characteristics of the Hadamard and Walsh matrix where the unused integer sign values will be removed. The test results show that all methods used have the same level of imperceptibility. In terms of embedding time, the Hadamard and Walsh-based method reduced the average time by 9% and 16% respectively compared to the singular value-based method, while the proposed method was able to achieve 20%. Similar results were also obtained during authentication where the Hadamard and Walsh method increased the speed by 9% and 18% while the proposed method was able to reach 20%. The proposed method is able to achieve the highest speed while maintaining the level of imperceptibility.

Keywords—image authentication, tampering, Hadamard matrix, Walsh matrix, unsigned integer

I. INTRODUCTION

Cloud storage systems are growing rapidly, encouraging people to upload image data to cloud storage, but this also raises issues related to the integrity and confidentiality of the data during the transmission process [1]. The development of image manipulation technology makes the image processing process faster and easier. This makes manipulated imagery difficult to distinguish from the original image with the naked eye [2]. This requires the development of image authentication technology that is able to detect tampering in certain regions of an image [3].

Digital watermarking is the most appropriate solution to solve the problem of manipulation on any type of data because it can recognize data manipulation that occurs in it accurately compared to passive detection techniques such as [4]. In general, there are two types of watermarking, namely robust watermarking which aims to protect the copyright of a data while fragile watermarking is used in the data authentication process. Research on fragile watermarking has grown rapidly

in recent years [5] not only on common images commonly shared by the public, but also in the medical world to diversify Computed Tomography (CT) imagery [6], Information Authentication and Integrity (IAI) based Block Chain [7], to Deep Neural Network (DNN)-based architecture protection [8], [9].

Fragile watermarking is developed rapidly and widely applied, requires it to have precise detection capabilities, a high level of imperceptibility, and fast computational time [10], [11] especially in large datasets that use distributed computing [12]. Recent studies have used a variety of techniques to increase velocity from a simple techniques on spatial domains based on the most and least significant bits [13], combinations of inpainting and bilateral filters on discrete wavelet transformation (DWT) domains [14], until the use of complex DNN architectures [15]. Methods based on transformation and decomposition do have the most ideal performance because they have good detection and imperceptibility capabilities with fast computational times [16], thus making matrix decomposition methods develop very rapidly.

Some popular decomposition methods are those based on QR values [17] and singular matrices such SVD [18], [19]. Matrix decomposition-based methods have a good level of robustness, acceptability, and detection capabilities but have high complexity. Other studies have tried to solve this problem by adding other algorithms such as logistic map [20] and Block Truncation Coding (BTC) [21]. Another approach is done by replacing the singular value generation process by using signed cosine matrix [22], Hadamard Matrix [23], and Walsh Matrix [24] in order to get faster computational times.

Singular values can be obtained by various decomposition techniques but basically the way these methods work is to get the highest value of the singular matrix by eliminating unnecessary processes. The Hadamard and Walsh matrices have similar properties on the largest value with faster running time. However, they still have useless signed value. Therefore, this study proposes a matrix generation model that is more optimal compared to previous methods using the combination of Hadamard and Walsh matrices to eliminate unnecessary values and get the advantages of both algorithms. The proposed model is based on unsigned integer values with fewer bits compared to both matrix-based singular and signed integers.

II. METHODS

A. Existing Methods

One of the most popular singular based method was proposed by [18] by dividing the largest value of the singular matrix S into a grouped block. The singular value is generated from the 7-MSB. SVD is very powerful in maintaining the visual quality or imperceptibility, but it has large computational cost due to the complexity of the decomposition processes as shown in (1) until (3):

$$SVD(B) = [U S V] \quad (1)$$

$$U = BVS^T \quad (2)$$

$$(B^T B - \lambda I) = 1/V \quad (3)$$

Where B is 7-MSB of image block, U and V is pair of orthogonal matrices, S is the singular matrix that is generated from the solution of determinant of $(B^T B - \lambda I)$, T is transpose of a matrix and I is identity matrix. SVD has high complexity due to the unnecessary generation of orthogonal matrices of U and V , hence several studies have to solve this problem.

A study from [23] proposed a solution to reduce the complexity of the singular based model using a Hadamard matrix which contains a signed integer number between -1 and 1. It is successfully reduce the complexity of SVD using a simpler matrix which have a close value to S matrix as shown in (4) and (5):

$$H_4 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \otimes \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \quad (4)$$

$$H * B * H = Q \quad (5)$$

Where H is Hadamard matrix. The first and largest element is close to largest singular value $Q(1) \approx S(1)$. The similar approach is used in [24] by partitioning the Walsh matrix W . Equation (6) shows the descending ordering of the sign change of [0 3 1 2] from the Hadamard matrix. Hence, it works like the Hadamard block as shown in (7) and (8):

$$W_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 \end{bmatrix} \quad (6)$$

$$W * B * W = R \quad (7)$$

$$Q(1) = R(1) \quad (8)$$

The difference between Walsh and Hadamard is the ordered of the unsigned value from the second row nor column. The sequence change also serves to replace the Arnold Scramble used in [18] and [23]. Hence the use of Walsh matrix is able to work faster than SVD and Hadamard. However, both Hadamard and Walsh use a signed integer matrix within size of 2^n , whereas the most important part is the first row and column. Hence, this paper proposes a new matrix that generates from Walsh and Hadamard elements in order to remove the unnecessary unsigned values.

B. Proposed method

This study uses the same characteristics of the Walsh and Hadamard matrices to form a new matrix that is more efficient in the process of embedding and extracting watermarks with the following steps:

1. Read the host image and then split it into block of image B with size of 4×4

2. Set all of the least significant bit of B into 0

$$B' = B - (B \bmod 2) \quad (9)$$

3. Cast the H'_4 and W_4 into unsigned integer to trim the unsigned value H'_4 and W'_4 respectively:

$$H'_4 = \text{uint}(H_4) \quad (10)$$

$$W'_4 = \text{uint}(W_4) \quad (11)$$

where the minimum length of unsigned integer is 2.

4. Build the new matrix M_4 using logical AND operator as follows:

$$M_4 = H'_4 \wedge W'_4 \quad (12)$$

5. Generate new matrix S for embedding processes through M :

$$S = \frac{M_4 * B' * M_4}{4} \quad (13)$$

6. Get the first element of S which is the largest or $S(1)$.
7. The following step for embedding and extraction are identical with [18] start from the stage 6 until 22 which is used in [23] as well.

III. RESULTS AND DISCUSSIONS

This experiment is done using 12th Gen Core i7 @2.1 GHz with LPDDR5 16GB @4800MHz on Windows 11. All algorithms are run on the same machine to get a fair comparison. The dataset was selected randomly from Kodak Lossless True Color Image Suit of 6 images with *.png extension.

A. Image Quality

The first measurement is quality or imperceptibility which measure the difference between host image and the watermarked image. This experiment uses the Peak Signal to Noise Ratio (PSNR) and Structural Similarity (SSIM) measurements which are the most widely used in the field of watermarking in order to obtain more comprehensive results. Table I shows that the results of all the methods used show an average value of PSNR of 55.9. The test results using SSIM in Table II also show the same thing with the average value of all methods is 0.9996.

Similar results are due to all methods using the same basic concept, namely the largest value of a singular block as referred to in (1), which can be approximated by the Hadamard-based method with a range of values [-1, 1] in (5). Another approach that have the value is the Walsh-based matrix as shown in (8). Both methods are used in this paper to form a new matrix that has the same value in the range [0 1] as seen in (12). The proposed method produces a close value to S so as to produce a very similar level of imperceptibility.

TABLE I. COMPARISON OF IMAGE QUALITY IN PSNR

Image	SVD[18]	Hadamard[23]	Walsh[24]	Proposed
Image02	55.91	55.91	55.92	55.92
Image07	55.89	55.90	55.91	55.91
Image08	55.89	55.91	55.91	55.91
Image12	55.90	55.92	55.92	55.92
Image19	55.90	55.91	55.91	55.91
Image23	55.88	55.89	55.88	55.88

TABLE II. COMPARISON OF IMAGE QUALITY IN SSIM

Image	SVD[18]	Hadamard[23]	Walsh[24]	Proposed
Image02	0.9999	0.9999	0.9999	0.9999
Image07	0.9996	0.9996	0.9996	0.9996
Image08	0.9997	0.9997	0.9997	0.9997
Image12	0.9997	0.9997	0.9997	0.9997
Image19	0.9993	0.9993	0.9993	0.9993
Image23	0.9996	0.9996	0.9996	0.9996



Fig. 1. Tampering on images; left: original image; center: manipulated image; right: tampering detection

B. Authentication

This section will discuss visual authentication testing as shown in Figure 1. The test will begin by manipulating images through 6 types of schemes namely moving, copy, enhancement, removal, addition or copy move and transformation. The first manipulation performed on Image02 is the process of moving by duplicating the door handle object to the right and removing the original object. The authentication results show that there have been changes in 2 areas, namely the position of the original object being deleted and the new object being added.

The next manipulation process is duplicating the pink hibiscus object on Image07 which can also be detected as a new object. The third process is enhancement of the moss-colored house object by adjusting the color hue to reddish. The results of the third test on Image08 show that there is a change in the overall object. Image12 undergoes a process of disappearance on the couple object on the beach. This object is also used in the process of adding to Image19. The authentication process from both shows changes with the same object shape. The final tampering process is carried out by carrying out a transformation in the form of a horizontal flip on the yellow-green parrot object which produces a symmetrical object indicating a change in the object's direction on Image23.

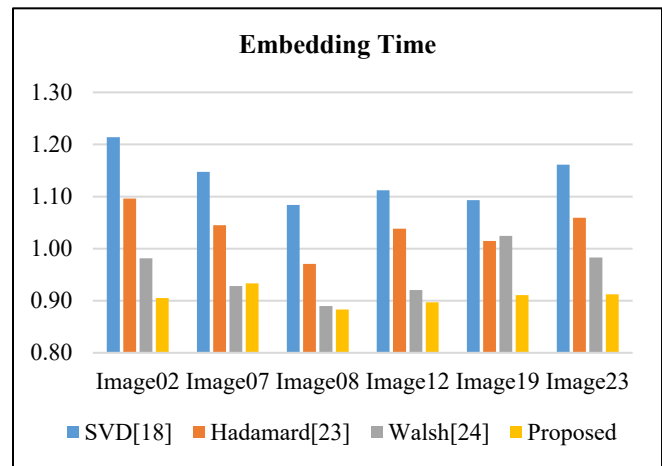


Fig. 2. Comparison of embedding time in second(s)

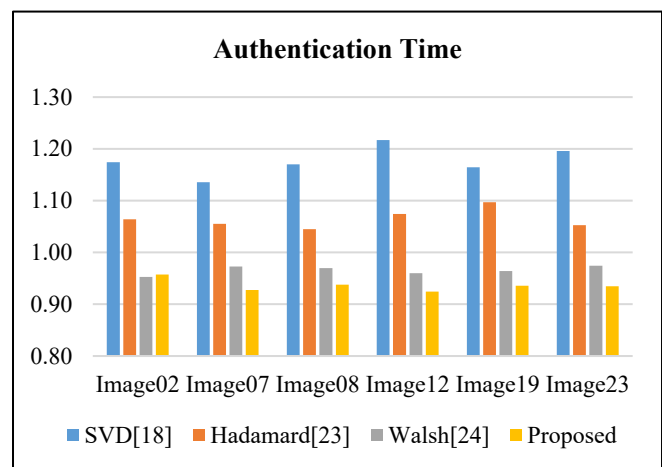


Fig. 3. Comparison of authentication time in second(s)

C. Computational Time

This experiment was conducted to determine the duration of the computational time required for each method. Testing is done on the process of embedding and extraction or authentication. A comparison of these tests can be seen in Figures 2 and 3. The test results show that the Hadamard and Walsh signed matrix-based method is able to work faster on average by 9% and 16% respectively. Meanwhile, the proposed method can accelerate the computational process by an average of 20% in the embedding process. The same results were also seen in the authentication process where the average reduction in computing time from the Hadamard and Walsh method was 9% and 18%, while the proposed method was able to increase the average speed by 20%.

The proposed method uses the characteristics of the two previous methods which have values -1 and 1 to form the largest value of the image block. They can be simplified into a matrix that has a value range of 0 and 1. This simplification can be done because the Walsh and Hadamard matrix has an inversely proportional integer sign as shown in (4) and (6). These characteristics can be used to form a new matrix that can change the value -1 to 0 so as to simplify the computational process while maintaining the original function shown in (12) and (13).

IV. CONCLUSION

This article proposes the formation of a new matrix as a method to reduce processing time on previous methods based on singular value and signed integer matrices. Matrix generation is carried out based on the characteristics of the Hadamard and Walsh matrices which have opposite integer signs in addition to the first row and column. The results of image quality testing show that all methods have the same level of imperceptibility. Tests were also carried out to measure computation time which showed that the Hadamard and Walsh matrix was able to increase the average speed of the embedding and authentication processes by 9% and 17%. Meanwhile, the proposed method can increase the average speed by 20%. It shows that the proposed method is able to reduce computational costs while maintaining imperceptibility.

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REFERENCES

- [1] L. Huang, D. Kuang, C. long Li, Y. jian Zhuang, S. hua Duan, and X. yi Zhou, "A self-embedding secure fragile watermarking scheme with high quality recovery," *J. Vis. Commun. Image Represent.*, vol. 83, no. January, p. 103437, 2022, doi: 10.1016/j.jvcir.2022.103437.
- [2] M. Ernawati, F. Ernawan, Y. Abbker, M. Fakhreldin, and P. W. Adi, "Image Splicing Forgery Approaches: A Review and Future Direction," in *2022 6th International Conference on Informatics and Computational Sciences (ICICoS)*, IEEE, Sep. 2022, pp. 134–139. doi: 10.1109/ICICoS56336.2022.9930543.
- [3] P. Lefèvre, P. Carré, C. Fontaine, P. Gaborit, and J. Huang, "Efficient image tampering localization using semi-fragile watermarking and error control codes," *Signal Processing*, vol. 190, 2022, doi: 10.1016/j.sigpro.2021.108342.
- [4] J. Nirmal Jothi and S. Letitia, "Tampering detection using hybrid local and global features in wavelet-transformed space with digital images," *Soft Comput.*, vol. 24, no. 7, pp. 5427–5443, 2020, doi: 10.1007/s00500-019-04298-4.
- [5] V. Sisaudia and V. P. Vishwakarma, "A secure gray-scale image watermarking technique in fractional DCT domain using zig-zag scrambling," *J. Inf. Secur. Appl.*, vol. 69, no. August, p. 103296, 2022, doi: 10.1016/j.jisa.2022.103296.
- [6] S. Shivani, "Verifiable medical images for E-healthcare: A novel watermarking approach using robust bit-wise association of self-mutating offsprings of pixels," *Microprocess. Microsyst.*, vol. 90, no. December 2021, 2022, doi: 10.1016/j.micpro.2022.104483.
- [7] J. Lou and W. Lu, "Construction information authentication and integrity using blockchain-oriented watermarking techniques," *Autom. Constr.*, vol. 143, no. January, p. 104570, 2022, doi: 10.1016/j.autcon.2022.104570.
- [8] G. Zhao, C. Qin, H. Yao, and Y. Han, "DNN self-embedding watermarking: Towards tampering detection and parameter recovery for deep neural network," *Pattern Recognit. Lett.*, vol. 164, pp. 16–22, 2022, doi: 10.1016/j.patrec.2022.10.013.
- [9] M. Botta, D. Cavagnino, and R. Esposito, "NeuNAC: A novel fragile watermarking algorithm for integrity protection of neural networks," *Inf. Sci. (Ny)*, vol. 576, pp. 228–241, 2021, doi: 10.1016/j.ins.2021.06.073.
- [10] P. W. Adi and P. Arsiwi, "A novel watermarking method using hadamard matrix quantization," *J. ICT Res. Appl.*, vol. 14, no. 1, pp. 1–15, 2020, doi: 10.5614/itbj.ict.res.appl.2020.14.1.1.
- [11] B. Bolourian Haghighi, A. H. Taherinia, and A. H. Mohajerzadeh, "TRLG: Fragile blind quad watermarking for image tamper detection and recovery by providing compact digests with optimized quality using LWT and GA," *Inf. Sci. (Ny)*, vol. 486, pp. 204–230, 2019, doi: 10.1016/j.ins.2019.02.055.
- [12] S. Rani and R. Halder, "An efficient format-independent watermarking framework for large-scale data sets," *Expert Syst. Appl.*, vol. 208, no. July, p. 118085, 2022, doi: 10.1016/j.eswa.2022.118085.
- [13] S. Bravo-Solorio, F. Calderon, C. T. Li, and A. K. Nandi, "Fast fragile watermark embedding and iterative mechanism with high self-restoration performance," *Digit. Signal Process. A Rev. J.*, vol. 73, pp. 83–92, 2018, doi: 10.1016/j.dsp.2017.11.005.
- [14] H. M. Al-Otum and A. A. A. Ellubani, "Secure and effective color image tampering detection and self restoration using a dual watermarking approach☆," *Optik (Stuttg.)*, vol. 262, pp. 1–22, 2022, doi: 10.1016/j.ijleo.2022.169280.
- [15] S. Mellimi, V. Rajput, I. A. Ansari, and C. W. Ahn, "A fast and efficient image watermarking scheme based on Deep Neural Network," *Pattern Recognit.*

- Lett.*, vol. 151, pp. 222–228, 2021, doi: 10.1016/j.patrec.2021.08.015.
- [16] M. Begum, J. Ferdush, and M. S. Uddin, “A Hybrid robust watermarking system based on discrete cosine transform, discrete wavelet transform, and singular value decomposition,” *J. King Saud Univ. - Comput. Inf. Sci.*, vol. 34, no. 8, pp. 5856–5867, 2022, doi: 10.1016/j.jksuci.2021.07.012.
- [17] Y. Chen, Z. G. Jia, Y. Peng, Y. X. Peng, and D. Zhang, “A new structure-preserving quaternion QR decomposition method for color image blind watermarking,” *Signal Processing*, vol. 185, p. 108088, 2021, doi: 10.1016/j.sigpro.2021.108088.
- [18] Q. Kang, K. Li, and H. Chen, “An SVD-based Fragile Watermarking Scheme With Grouped Blocks,” in *International Conference on Information Technology and Electronic Commerce*, 2014, pp. 172–179.
- [19] N. Zermi, A. Khaldi, R. Kafi, F. Kahlessenane, and S. Euschi, “A DWT-SVD based robust digital watermarking for medical image security,” *Forensic Sci. Int.*, vol. 320, 2021, doi: 10.1016/j.forsciint.2021.110691.
- [20] N. R. Neena and R. Shreelekshmi, “Fragile watermarking scheme for tamper localization in images using logistic map and singular value decomposition,” *J. Vis. Commun. Image Represent.*, vol. 85, no. April, p. 103500, 2022, doi: 10.1016/j.jvcir.2022.103500.
- [21] M. Swain and D. Swain, “An effective watermarking technique using BTC and SVD for image authentication and quality recovery,” *Integration*, vol. 83, no. June 2021, pp. 12–23, 2022, doi: 10.1016/j.vlsi.2021.11.004.
- [22] F. Ernawan, P. W. Adi, S. C. Liew, E. A. Sarwoko, and E. Winarno, “Fast image watermarking based on signum of cosine matrix,” *Indones. J. Electr. Eng. Comput. Sci.*, vol. 25, no. 3, pp. 1383–1391, 2022, doi: 10.11591/ijeecs.v25.i3.pp1383-1391.
- [23] P. W. Adi, R. Kusumaningrum, and Khadijah, “Active Tampering Detection for Image Authentication using Hadamard Tracing,” in *2022 International Conference on Electrical Engineering and Informatics (ICELTICs)*, IEEE, Sep. 2022, pp. 78–82. doi: 10.1109/ICELTICs56128.2022.9932130.
- [24] P. W. Adi and P. Arsiwi, “Fast and Robust Watermarking Method using Walsh Matrix Partition,” *2019 2nd Int. Semin. Res. Inf. Technol. Intell. Syst. ISRITI 2019*, pp. 468–472, 2019, doi: 10.1109/ISRITI48646.2019.9034627.